Radiomics Study Report: Neuroblastoma Classification

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Executive Summary

This radiomics study aimed to classify medical images as either 'healthy' or 'neuroblastoma' using radiomic features extracted from segmented images. The analysis utilized 13 samples with 10 key radiomic features. The machine learning model achieved modest performance with mean accuracy of 60% (SD: 20%), AUC of 40% (SD: 20%), and F1-score of 46.7% (SD: 26.7%).

1 Methodology

1.1 Dataset Characteristics

- **Sample Size**: 13 medical images
- **Class Distribution**: Balanced (healthy vs neuroblastoma)
- **Segmentation**: Completed using automated segmentation agent
- **Labels**: Label 1 (neuroblastoma), Label 255 (healthy tissue)

1.2 Feature Extraction

Radiomic features were extracted using PyRadiomics library:

- **Total Features Extracted**: 107 features per sample
- **Feature Classes**: First-order, Shape, GLCM, GLRLM, GLSZM, GLDM, NGTDM
- **Key Feature Categories**: Texture, Shape, Intensity-based statistics

1.3 Feature Selection

Top 10 most important features identified through Random Forest importance analysis:

1. GLRLM RunVariance
2. GLDM DependenceNonUniformity

3. GLSZM HighGrayLevelZoneEmphasis
4. SurfaceVolumeRatio (Shape)
5. Flatness (Shape)
6. TotalEnergy (First-order)
7. Energy (First-order)
8. GLCM MCC
9. GLSZM LargeAreaEmphasis
10. GLRLM RunLengthNonUniformity

1.4 Machine Learning Modeling

- **Experiment Name**: classification_exp_20251218_184735
- **Cross-validation**: 5-fold cross-validation
- **Evaluation Metrics**: Accuracy, AUC, Precision, Recall, F1-score

2 Results

2.1 Model Performance Summary

Metric	Mean	Standard Deviation	Min	Max
Accuracy	60.0%	20.0%	50.0%	100.0%
AUC	40.0%	20.0%	0.0%	50.0%
Precision	53.3%	26.7%	33.3%	100.0%
Recall	53.3%	26.7%	33.3%	100.0%
F1-score	46.7%	26.7%	33.3%	100.0%

Table 1: Model Performance Metrics Across 5-Fold Cross-Validation

2.2 Key Findings

- The model demonstrated high variability across different cross-validation folds
- Texture features (GLRLM RunVariance, GLDM DependenceNonUniformity) showed the highest discriminative power
- Shape features (SurfaceVolumeRatio, Flatness) contributed moderately to classification
- First-order statistics (TotalEnergy, Energy) provided limited discriminative information

3 Discussion

3.1 Performance Evaluation

The developed classification model achieved suboptimal performance for clinical diagnostic applications:

- The AUC of 40% indicates poor discriminative ability
- High standard deviations across folds suggest model instability
- The limited sample size (n=13) likely contributed to performance limitations

3.2 Limitations

- **Sample Size**: Small dataset limiting model generalization
- **Feature Selection**: Limited to 10 features due to small sample size
- **Model Complexity**: Simple classifier due to data constraints
- **Cross-validation Variability**: High variability indicates model instability

4 Conclusion

The radiomics analysis for neuroblastoma classification using the current dataset and methodology demonstrated insufficient predictive performance for reliable clinical application. The key conclusions are:

- The hypothesis that radiomic features can effectively classify neuroblastoma cases is **rejected**
- Model performance metrics fall below acceptable thresholds for medical diagnostic applications
- Future studies should focus on larger datasets and more sophisticated modeling approaches

5 Recommendations

1. **Expand Dataset**: Increase sample size to improve model stability
2. **Feature Engineering**: Explore additional radiomic features and transformations
3. **Advanced Modeling**: Consider ensemble methods or deep learning approaches
4. **External Validation**: Validate findings on independent datasets
5. **Clinical Integration**: Consider integration with clinical biomarkers for improved performance

Technical Details

- **Software**: PyRadiomics for feature extraction, scikit-learn for modeling
- **Processing Time**: Feature extraction completed in 46.3 seconds
- **Output Files**: Model artifacts, performance metrics, and visualizations generated
- **Reproducibility**: Complete workflow documented in configuration files